

Chapter 12 and 13: Introduction to and Geometry of vectors

Worksheet

1. Are each of the following statements true or false?

- (a) There are scalars c_1, c_2 such that $c_1 \begin{bmatrix} 3 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} -5 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \end{bmatrix}$.
- (b) If c, d are scalars and $\mathbf{v}$ is a vector in $\mathbb{R}^2$, then $(c + d)\mathbf{v}$ is a vector in $\mathbb{R}^2$.
- (c) If c, d are scalars and $\mathbf{v}$ is a vector in $\mathbb{R}^2$, then $c\mathbf{v} + d$ is a vector in $\mathbb{R}^2$.
- (d) If $\mathbf{u}$ is a unit vector, so is $-\mathbf{u}$.
- (e) If $\mathbf{u}, \mathbf{v}$ are vectors in $\mathbb{R}^4$ with the norm of $\mathbf{u}$ equal to 4 and the norm of $\mathbf{v}$ equal to 10, then the maximum value that $\mathbf{u} \cdot \mathbf{v}$ can obtain is 40.

2. Consider $\mathbb{R}^3$, and the subset of vectors W in $\mathbb{R}^3$ which satisfy the following rule:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \text{ where } 2v_1 - 3v_2 + 4v_3 \geq 0.$$

In set notation, this given as: $W = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \mid 2v_1 - 3v_2 + 4v_3 \geq 0 \right\}$

- (a) Find an explicit vector $\mathbf{v}$ (that means using numbers in the coordinates) which satisfies the rule of W . We say then that $\mathbf{v}$ is an element of W , or $\mathbf{v} \in W$.
- (b) Find an explicit vector $\mathbf{w}$ (that means using numbers in the coordinates) which does NOT satisfy the rule of W . We say then that $\mathbf{w}$ is not an element of W , or $\mathbf{w} \notin W$.

3. Do there exist scalars a, b, c so that $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \begin{bmatrix} 7 \\ 4 \\ 14 \end{bmatrix}$, where $\mathbf{u} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix}$ and

$$\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} ?$$

4. For each of the following pair of vectors, find the distance and angle between them.

(a) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$

(b) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -4 \\ -5 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$.